HPRC 2026 • HUMAN PANGENOME REFERENCE CONSORTIUM

A Pangenome Sequence Search and Coordinate Translation Service for the UCSC Genome Browser

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The **UCSC Genome Browser** now speaks **pangenome**.

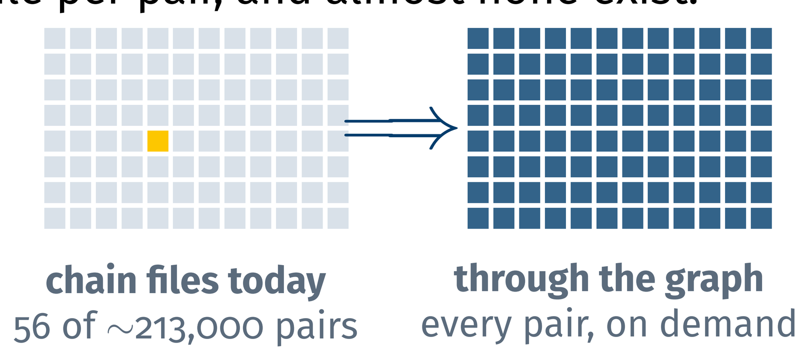
464 haplotypes scored per query

~20 ms gene-scale liftover

466 × 466 assembly pairs, on demand

01 The graph you couldn't ask

The HPRC v2.0 graph holds **233 samples and 464 haplotypes**. You can browse each assembly, but you cannot simply ask the graph a question: which haplotypes carry my sequence? Where does my region land on another assembly? Classic liftover needs a chain file per pair, and almost none exist:



We built that query layer, inside the UCSC Genome Browser.

02 What users get

Pangenome sequence search
BLAT, but for the whole pangenome: one search ranks all 464 haplotypes and links to the hit on any of them.

Any-to-any coordinate translation
Move a region between any two of the 466 assemblies, at interactive speed.

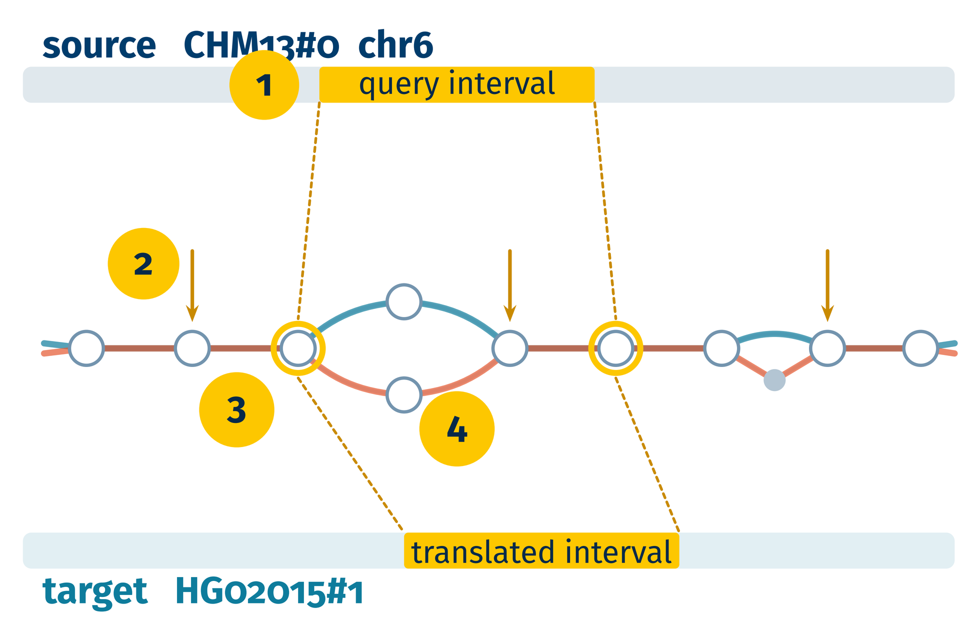
Annotation lifting
Arrive with the source's annotation tracks drawn at translated positions, plus an Alignment Differences track.

10 Acknowledgements

We thank our colleagues in the UC Santa Cruz Computational Genomics Lab, especially Adam Novak, Glenn Hickey, and Zia Truong, and the UCSC Genome Browser team. Built on `vg` and the HPRC v2.0 graph.

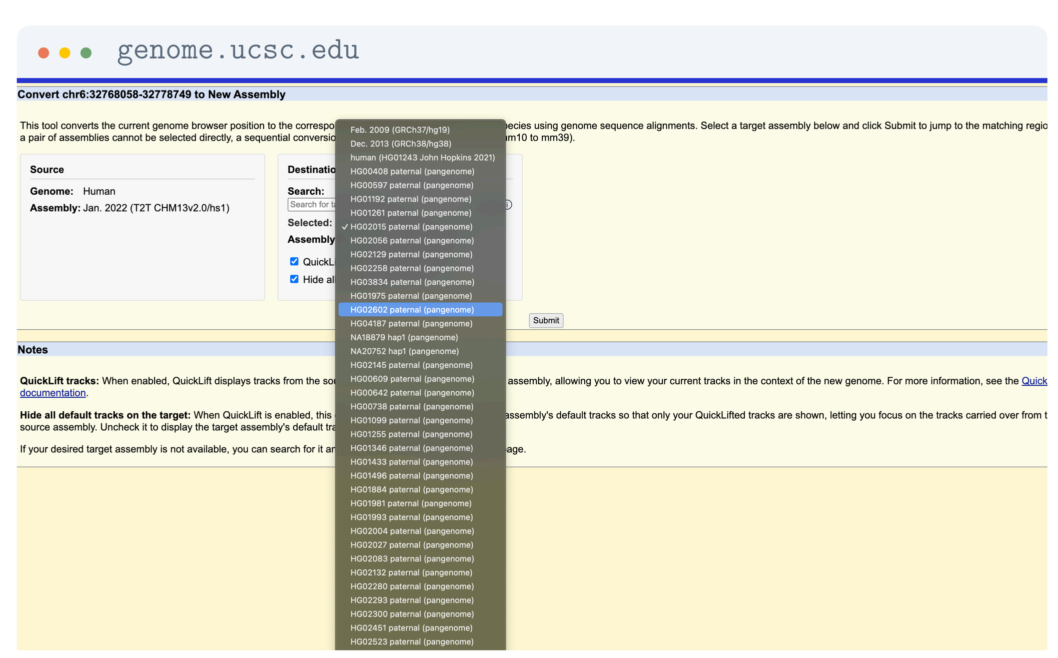
03 One index, every haplotype

Everything rides on our **tag-array index** (*lossless pangenome indexing*): it annotates each position of the BWT index with its graph position, so every match knows its node, and every node knows the haplotypes standing on it. Translation is a walk through the graph, not a lookup in 213,000 pairwise files:



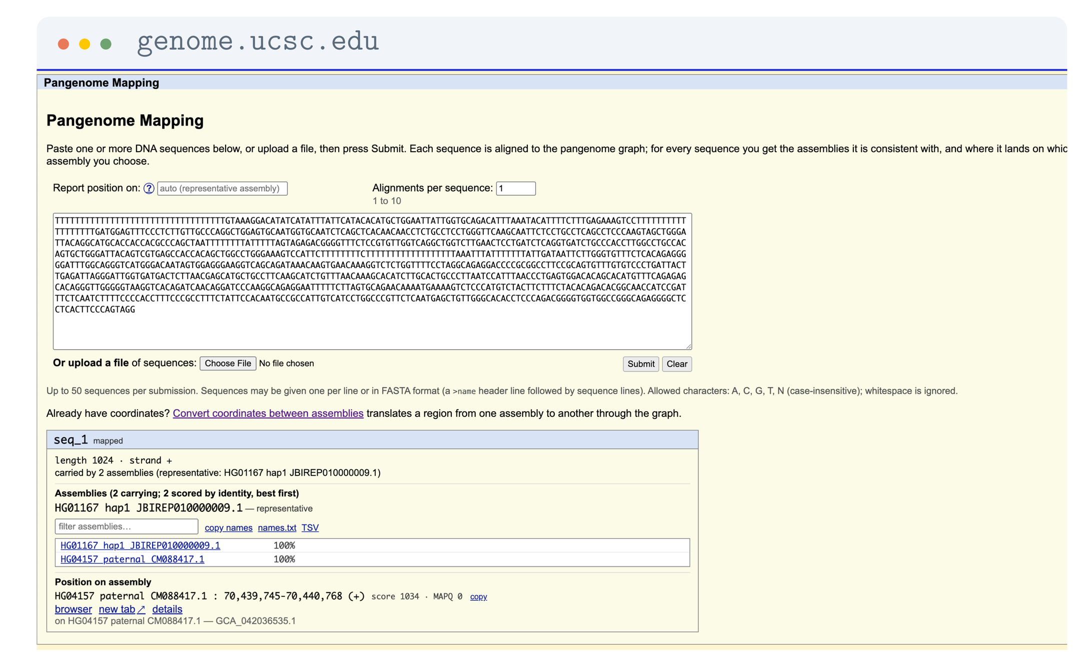
1 Locate the query interval's nodes on the source haplotype's path. **2** Tag arrays report, at every node, each haplotype standing on it and where. **3** Nodes visited exactly once by both source and target make guaranteed orthologous anchor pairs. **4** A walk between anchors emits per-base offsets, folded into chain blocks.

04 It fits the tools people know

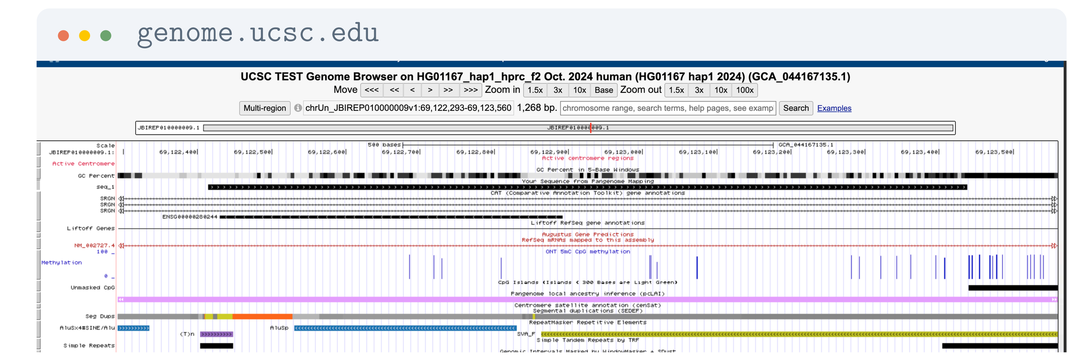


Pangenome assemblies appear right in the stock `hgConvert` menu; picking one dispatches the conversion through the graph.

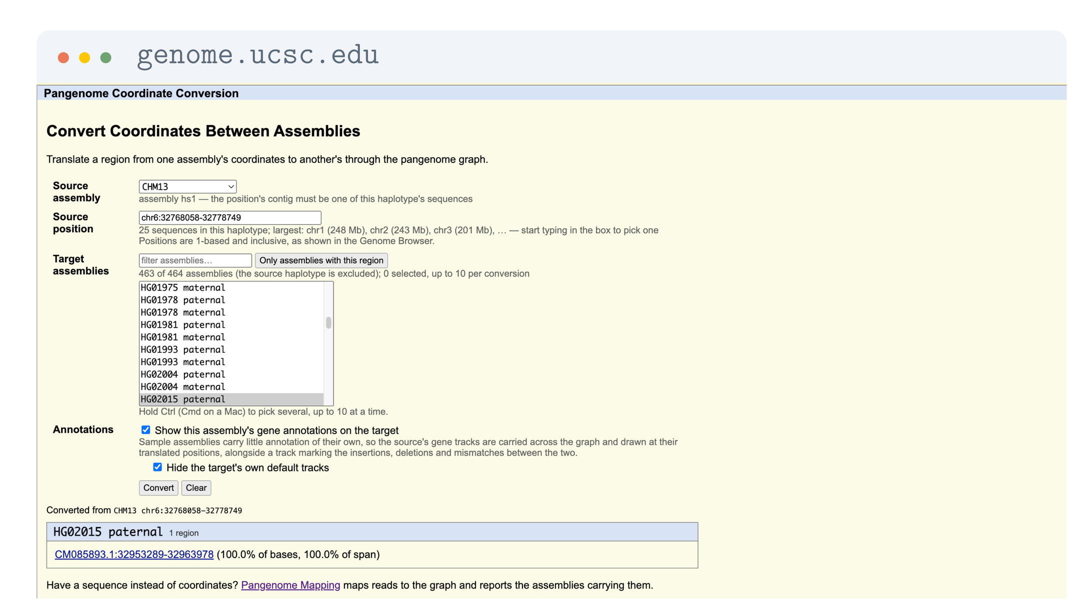
05 Search, in the browser



One search maps the sequence to the graph with `vg giraffe` and reports every assembly carrying it, ranked by identity. Picking a haplotype re-surjects the cached alignment, no remapping, and the hit lands as a native track with base-level differences:

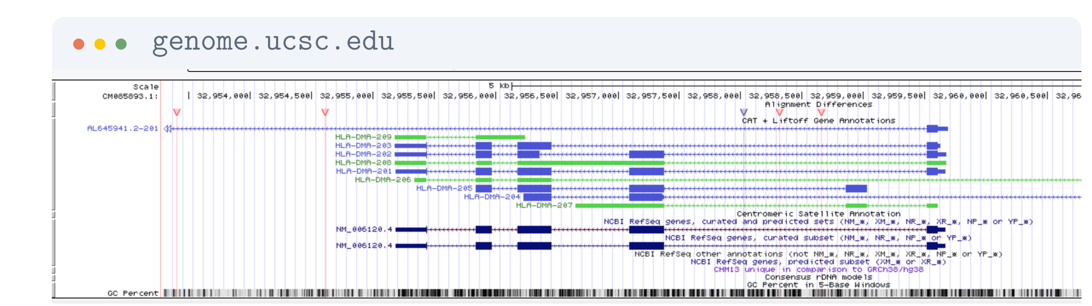


06 Convert between assemblies



CHM13 `chr6:32,768,058-32,778,749` translated to HG02015 paternal `CM085893.1:32,953,289-32,963,978`, covering 100% of bases. The picker can list only the assemblies that actually contain the region.

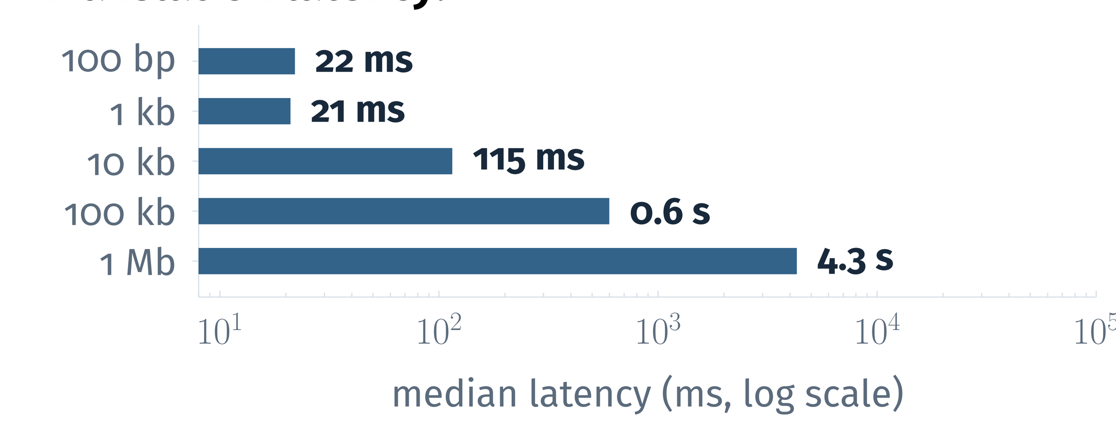
07 Land with annotations



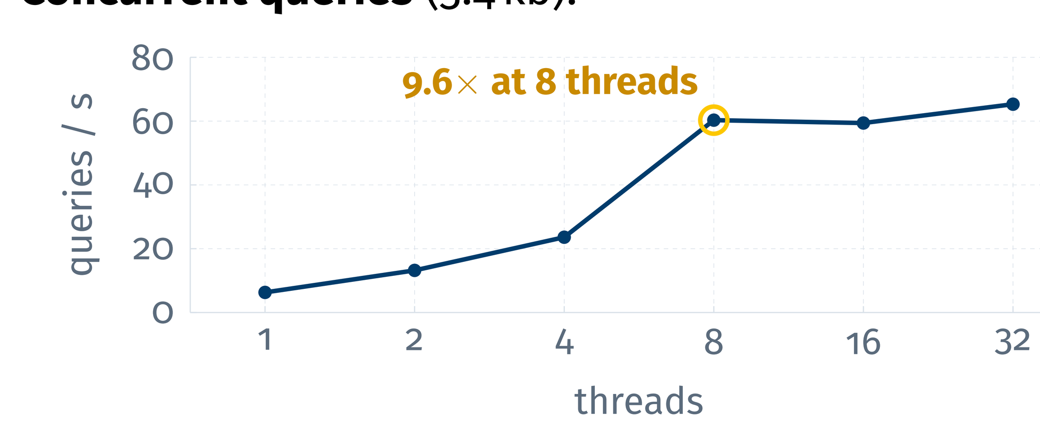
The converted region opens on HG02015 with annotations carried over from the source haplotype (here CHM13): CAT and Liftoff genes of the HLA-DMA cluster, RefSeq and CenSat, under an **Alignment Differences** track marking indels and mismatches.

08 Fast enough to be interactive

Translation latency:



Concurrent queries (3.4 kb):



Search, translation and annotation lifting all run at page-load speed.

09 What's next

Public release
Bringing pangenome sequence search and any-to-any coordinate translation to everyone, in the public UCSC Genome Browser.